UNIVERSITY OF SOUTH CAROLINA

ITEC 493: IT Security for Managers

Cloud Computing and Virtualization Security

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Module Objectives

By the end of this module, you should be able to:

1. Define the cloud and explain how it is used and managed

2. Explain virtualization

3. Describe cloud and virtualization security controls

4. List different secure network protocols

Introduction to Cloud Computing

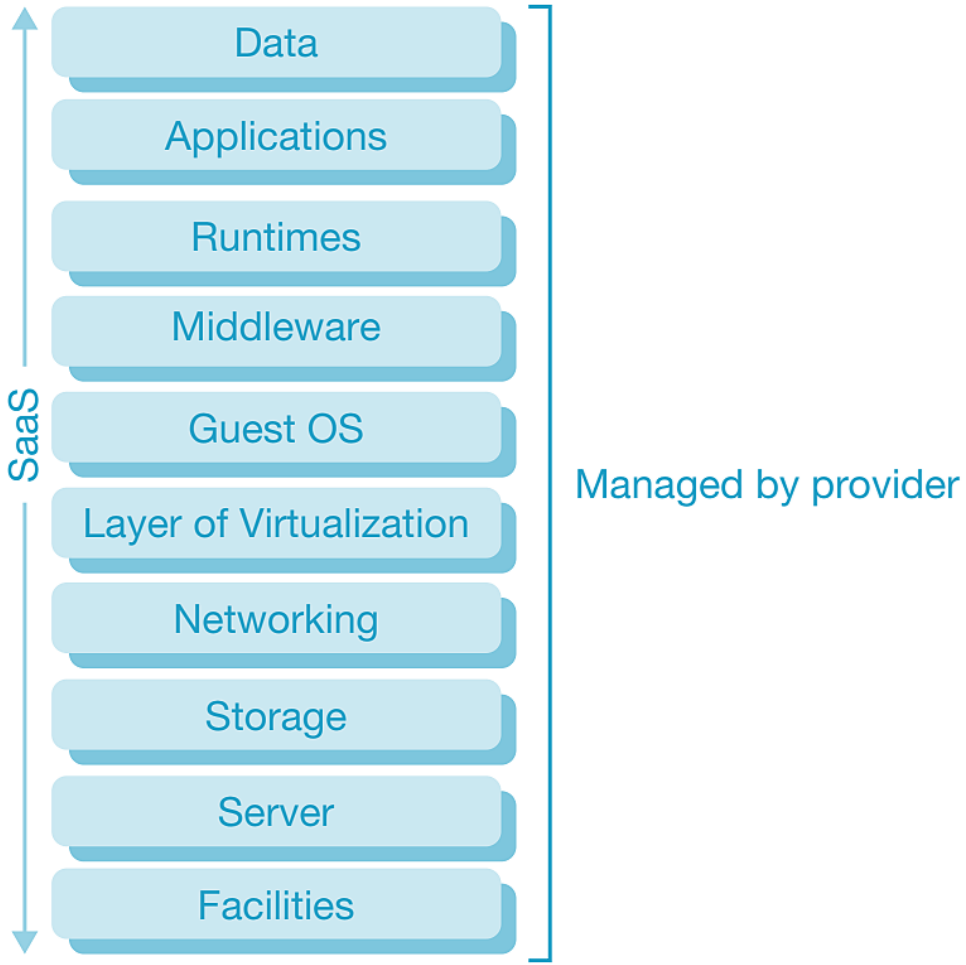
- In a *hosted services* environment, servers, storage, and the supporting networking infrastructure are shared by multiple enterprises over a remote network connection
- **Cloud computing** is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources
  - Entities that offer cloud computing are called **cloud service providers**
- The savings available through cloud computing are due to the following factors:
  - *Elasticity and scalability*
  - *Pay-per-use*
  - *On demand*
  - *Resiliency*

Introduction to Cloud Computing

- There are four service models in cloud computing:
  - **Software as a Service (SaaS)**
    - Vendor provides access to the vendor's software applications running on a cloud infrastructure
  - **Platform as a Service (PaaS)**
    - Consumers install and run their own specialized applications on the cloud computing network
  - **Infrastructure as a Service (IaaS)**
    - Vendor allows customers to deploy and run their own software, including OSs and applications
  - **Anything as a Service (XaaS)**
    - A broad category of subscription services related to cloud computing

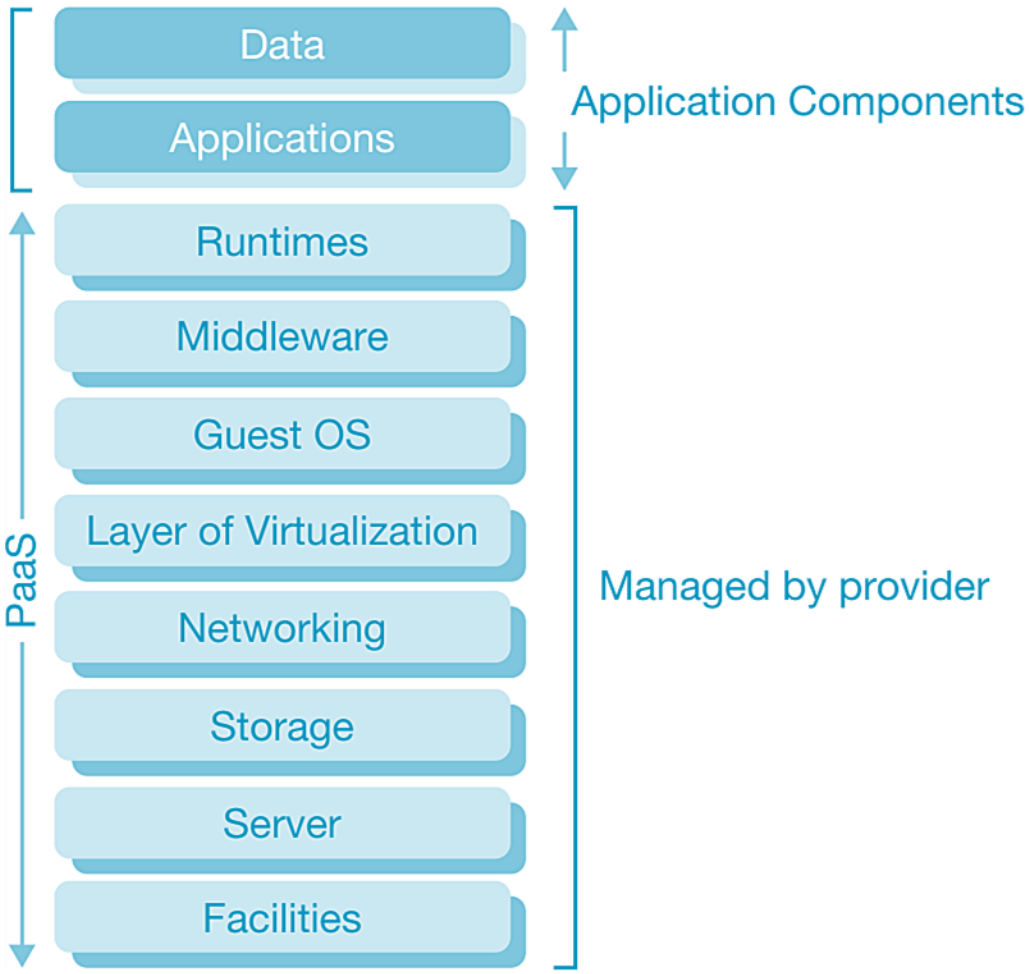
Software-as-a-Service

- Applications are supplied by the service provider
- The user does not manage or control the underlying cloud infrastructure or individual application capabilities
- Examples:
  - GoogleApps, Microsoft Office 365
  - Customer Relationship Management (CRM)
  - Enterprise Resource Planning (ERP)
  - Etc.



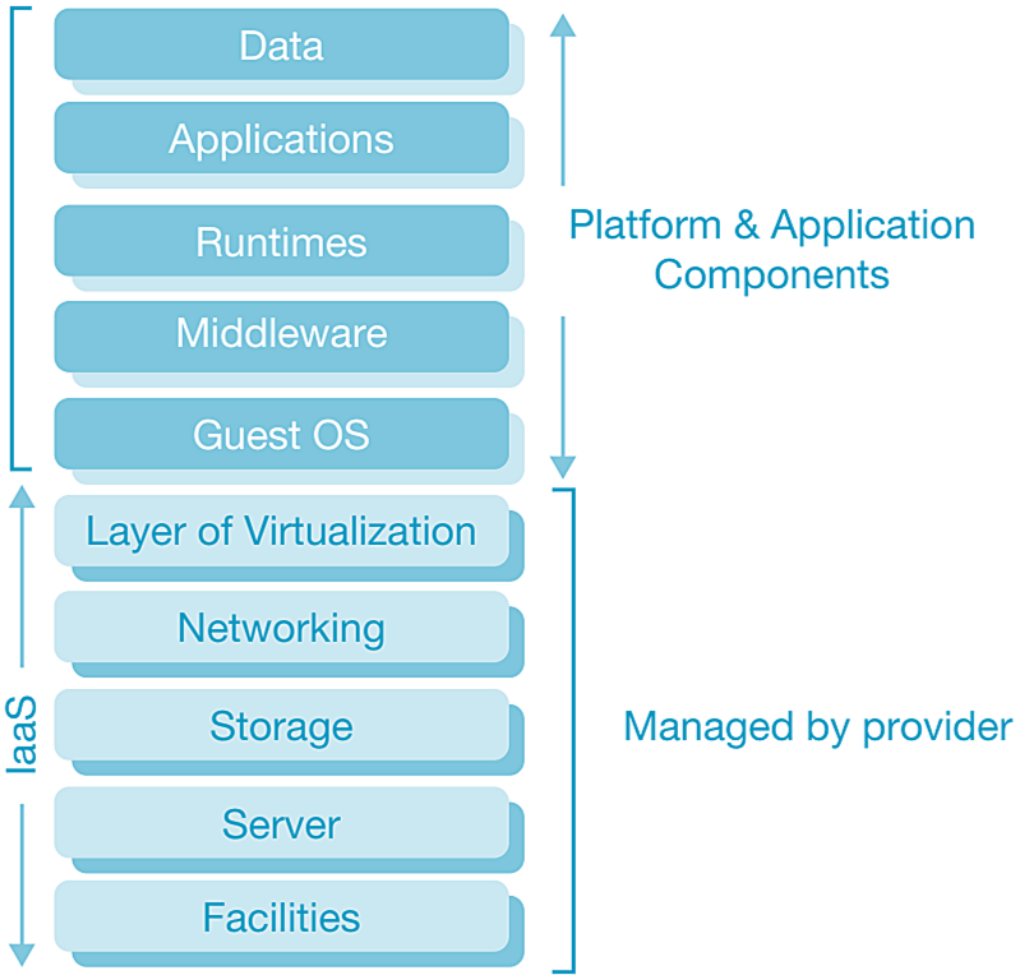
Platform-as-a-Service (PaaS)

- PaaS provides an environment for developers and companies to create, host and deploy applications
- Saves developers from the complexities of infrastructure
- Focus on writing code rather than provisioning services and infrastructure
- Not particularly useful when
  - The application must be portable.
  - Proprietary programming languages are used.
  - The hardware and software must be customized to improve the performance of the application

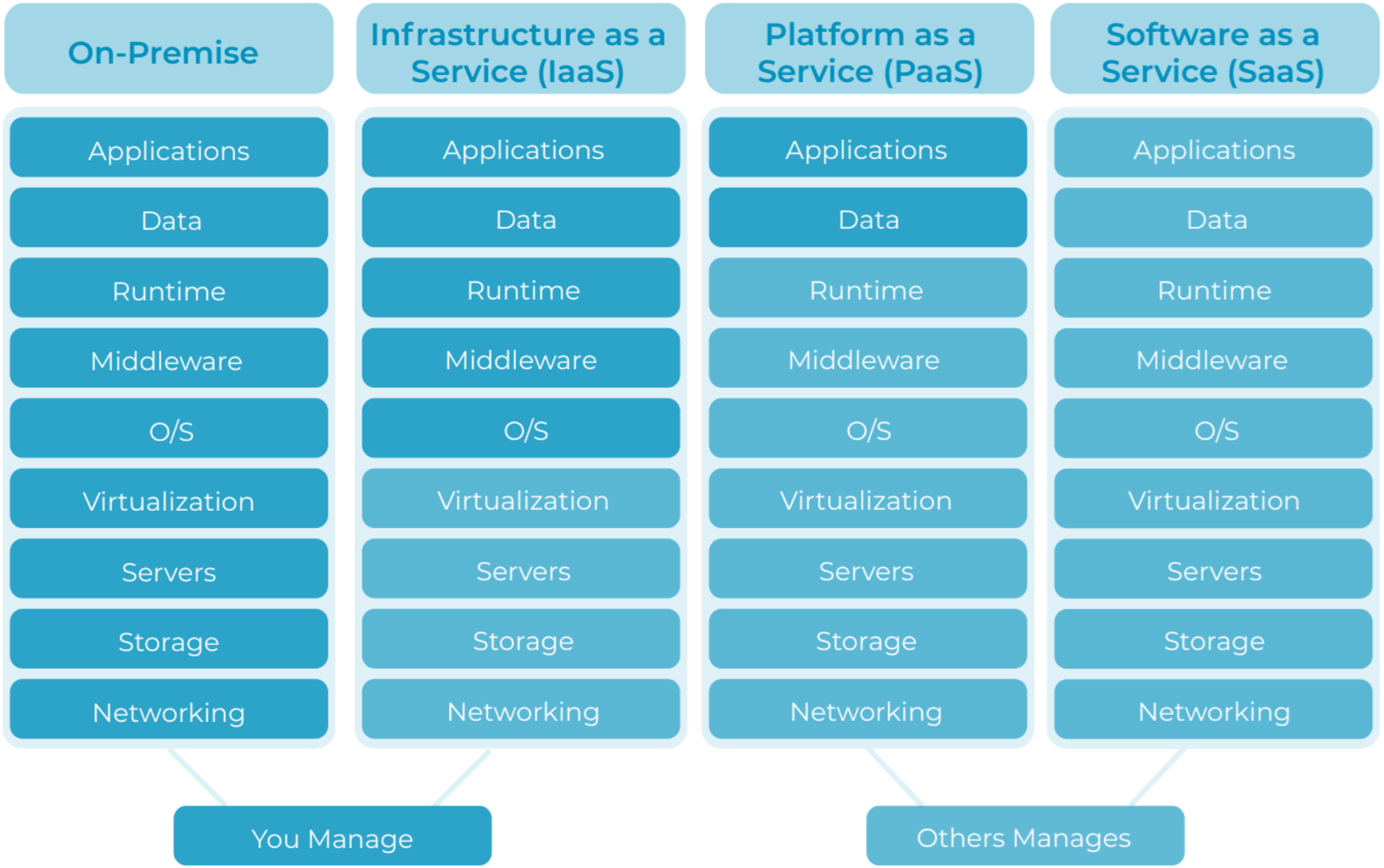


Infrastructure-as-a-Service (IaaS)

- The user does not manage or control the underlying cloud infrastructure
- But has control over operating systems, storage, deployed applications, and possibly limited control of some networking components, e.g., host firewalls
- Examples:
  - Amazon AWS EC2
  - Google Compute Engine



IaaS vs PaaS vs SaaS



Securing Cloud Computing

Security issue	Description
Unauthorized access to data	Improper cloud security configurations can result in data being left exposed.
Lack of visibility	Organizations have limited or no visibility into the security mechanisms of the cloud provider and thus cannot verify the effectiveness of security controls.
Insecure application program interfaces (APIs)	While APIs help cloud customers customize their PaaS by providing data recognition, access, and effective encryption, a vulnerable API can be exploited by threat actors.
Compliance regulations	Maintaining compliance requires that an organization know where its data is, who can access it, and how it is protected, but this can be difficult in an opaque cloud system, which lacks transparency.
System vulnerabilities	A cloud infrastructure is prone to system vulnerabilities due to complex networks and multiple third-party platforms.

Securing Cloud Computing

- Cloud Security Controls
  - Securing cloud computing involves using controls such as the following:
    - **Conducting audits** – a **cloud security audit** is an independent examination of cloud service controls
    - **Use Regions and Zones** – reliability and resiliency are achieved through duplicating processes across one or more geographical areas (called **high availability across zones**)
    - **Secrets management** – enables strong security and improved management of a microservices-based architecture, allowing the entire cloud infrastructure to remain flexible and scalable without sacrificing security

Securing Cloud Computing

- Application Security
  - One of an organization's security protections for cloud computing application security is to use a **cloud access security broker** (**CASB**)
  - CASB is a set of software tools or services that resides between the enterprises' on-premises infrastructure and the cloud provider's infrastructure
  - CASB acts as a "gatekeeper", ensuring that the security policies of the enterprise extend to its data in the cloud
- Security Virtual Device Solutions
  - A **next-generation secure web gateway** (**SWG**) examines both incoming and outgoing traffic and performs basic URL and monitoring in web applications
  - A **cloud firewall** is virtual software that functions in a similar manner to a physical security appliance by examining traffic into and out of the cloud

Virtual Network Security

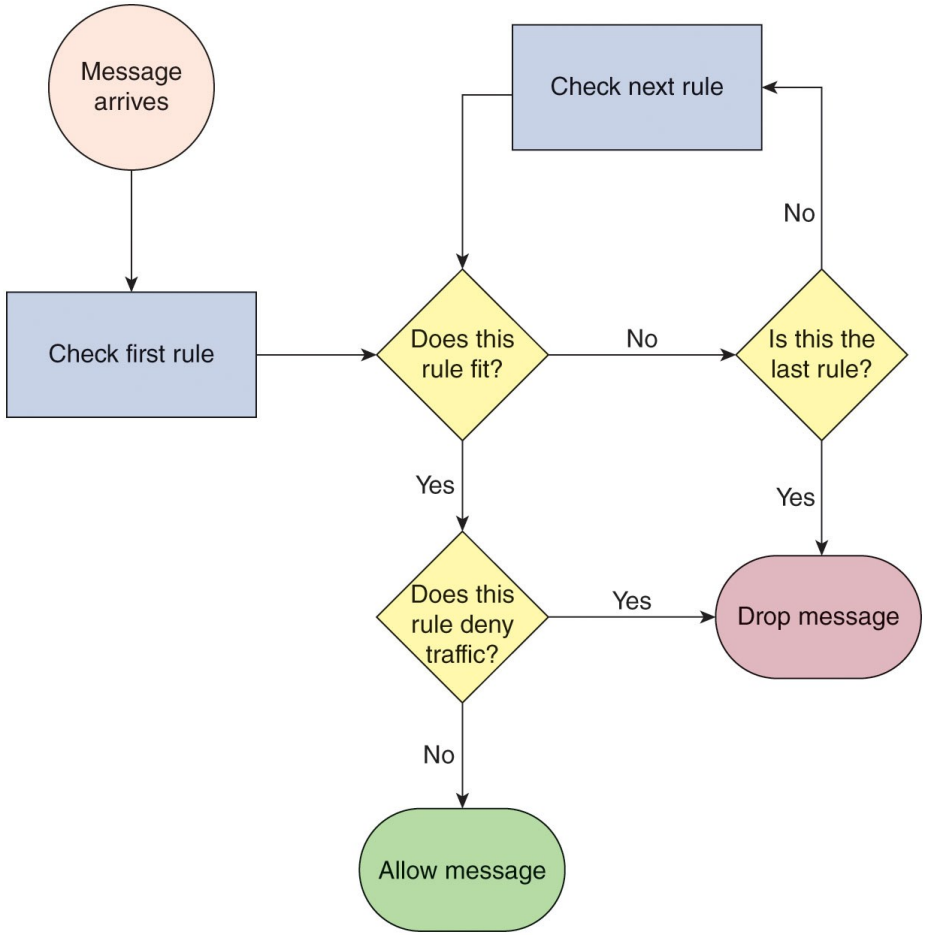
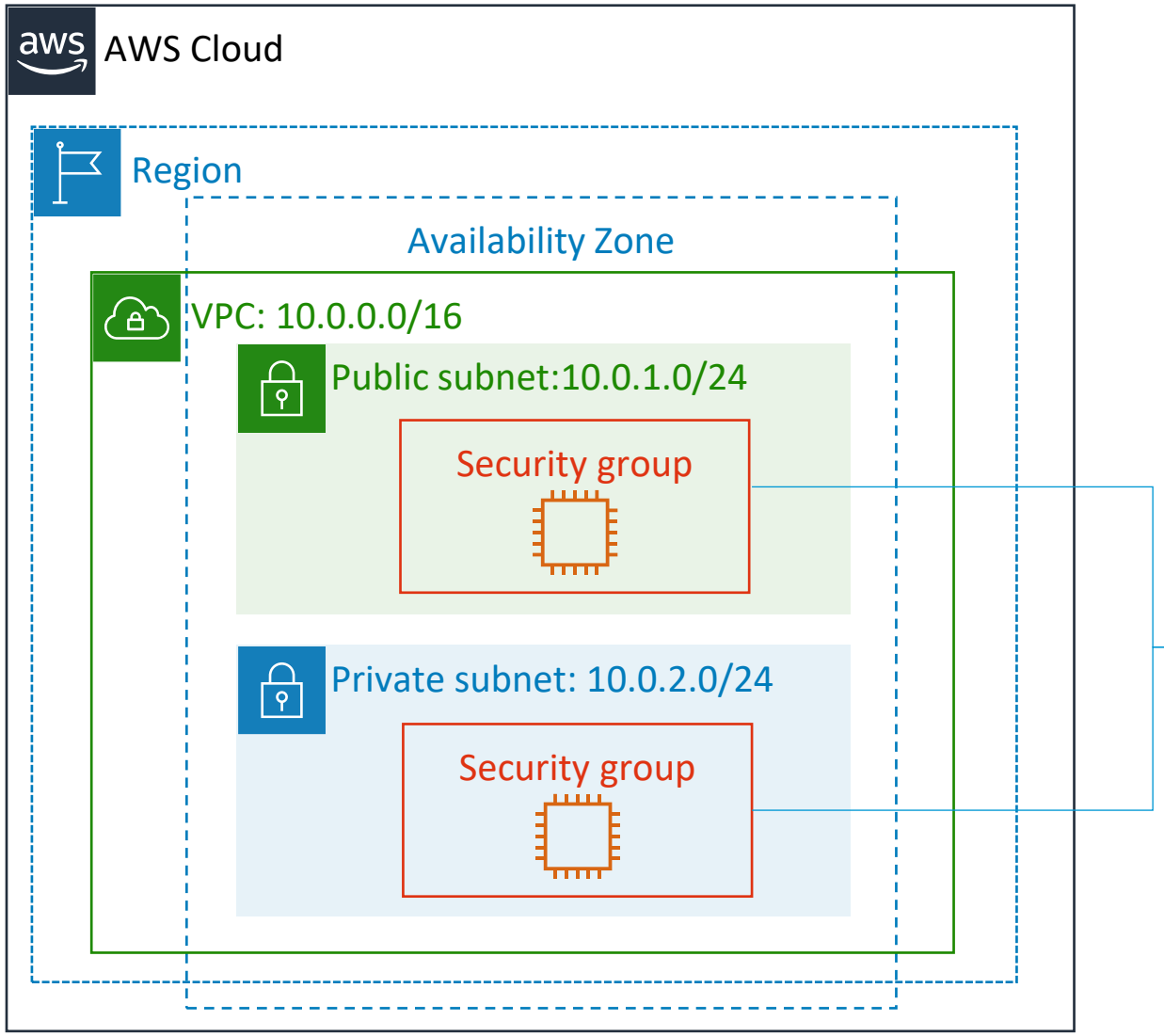


Figure 6-3 Firewall rules are checked in order of priority to determine whether a message is allowed to pass.

Security groups



Security groups act at the instance level.

Security groups

- Security groups have rules that control inbound and outbound instance traffic.
- Default security groups deny all inbound traffic and allow all outbound traffic.
- Security groups are stateful.

Inbound			
Source	**Protocol**	**Port Range**	**Description**
sg-*xxxxxxxx*	All	All	Allow inbound traffic from network interfaces assigned to the same security group.

Outbound			
Destination	Protocol	Port Range	Description
0.0.0.0/0	All	All	Allow all outbound IPv4 traffic.
::/0	All	All	Allow all outbound IPv6 traffic.

Custom security group examples

- You can **specify allow** rules, but not deny rules.
- **All rules are evaluated** before the decision to allow traffic.

Inbound			
Source	**Protocol**	**Port Range**	**Description**
0.0.0.0/0	TCP	80	Allow inbound HTTP access from all IPv4 addresses
0.0.0.0/0	TCP	443	Allow inbound HTTPS access from all IPv4 addresses
Your network's public IPv4 address range	TCP	22	Allow inbound SSH access to Linux instances from IPv4 IP addresses in your network (over the internet gateway)

Outbound			
Destination	Protocol	Port Range	Description
The ID of the security group for your Microsoft SQL Server database servers	TCP	1433	Allow outbound Microsoft SQL Server access to instances in the specified security group

Securing Cloud Computing

- Lack of a Cloud Conceptual Model
  - Physical networks use the Open Source Interconnection (OSI) seven-layer model to illustrate network functionality
  - The lack of a conceptual model like the OSI model makes selecting and managing security virtual devices more challenging
  - Different cloud-based conceptual models are starting to be proposed
    - However, no single model has been widely adapted
    - One model is shown in the next slide

Securing Cloud Computing

Layer and name	Description	Party responsible
5—Application Experience	End-user facing interface	Customer
4—Native Service	Create, store, process	Customer
3—Software-Defined Datacenter	Create infrastructure	SaaS—Cloud computing provider PaaS and IaaS—Customer
2—Virtualization Software	Software that virtualizes the hardware	Cloud computing provider
1—Physical Infrastructure	Buildings, power, cables, hardware, utilities	Cloud computing provider

Defining Virtualization

- What is Virtualization?
  - **Virtualization** is a means of managing and presenting computer resources without regard to physical layout or location
  - *Host virtualization* is a type of virtualization in which an entire operating system environment is simulated
  - A *virtual machine* (*VM*) is a simulated software-based emulation of a computer
  - The *host system* runs a hypervisor that manages the virtual operating systems and supports one or more guest systems
  - Virtualization is used to consolidate multiple physical servers into VMs that can run on a single physical computer

Virtualization Technologies (2 of 13)

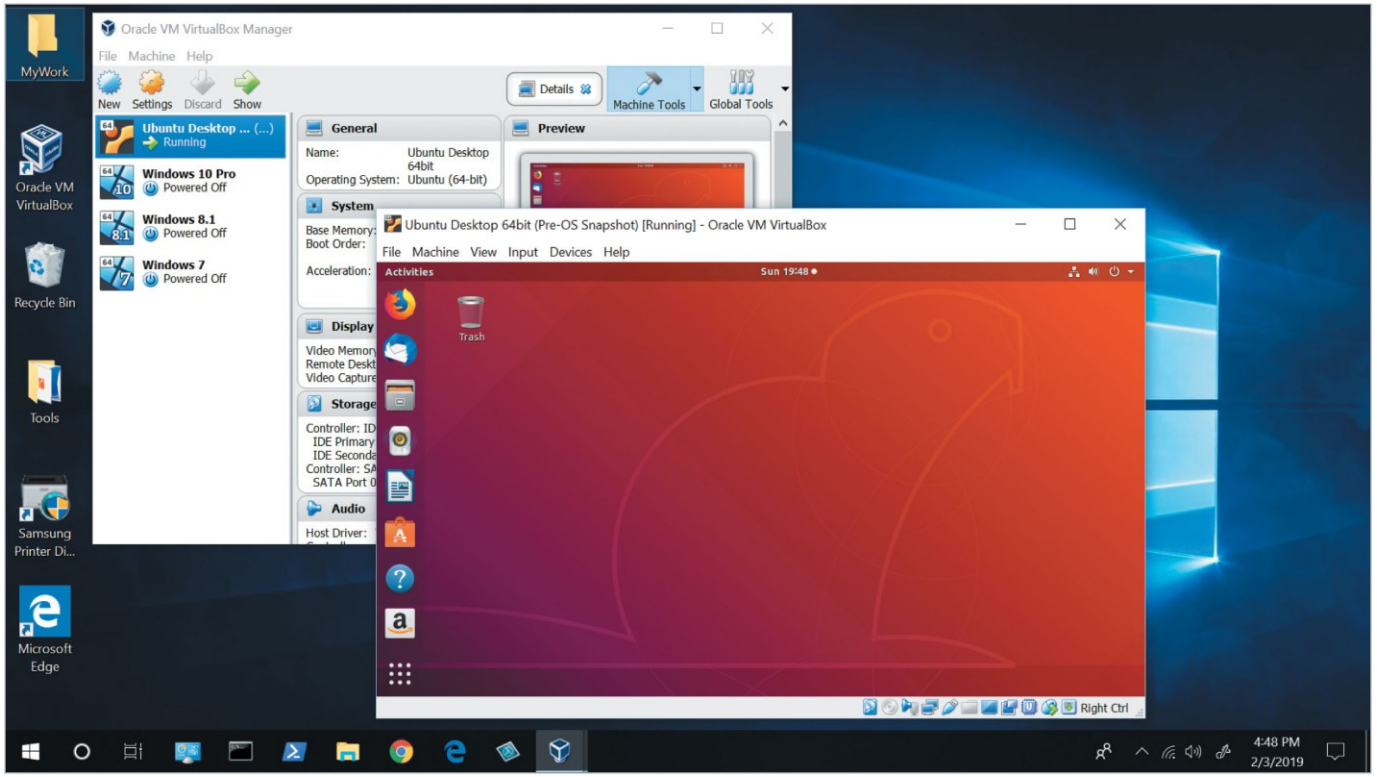


Figure 2-1 Windows 10 is installed on the physical machine while Ubuntu Desktop is installed in a VM running on the VirtualBox hypervisor.

Defining Virtualization

- The VM monitor program is called a *hypervisor*, which manages the VM operating systems
- Two types of hypervisor:
  - *Type I* – run directly on the computer's hardware instead of the underlying OS
  - *Type II* – run on the host OS, much like an application
- A **container** holds only the necessary OS components that are needed for that specific application to run
  - Reduces the necessary hard drive storage space and RAM needed
  - Allows for containers to start more quickly because the OS does not have to be started

Defining Virtualization

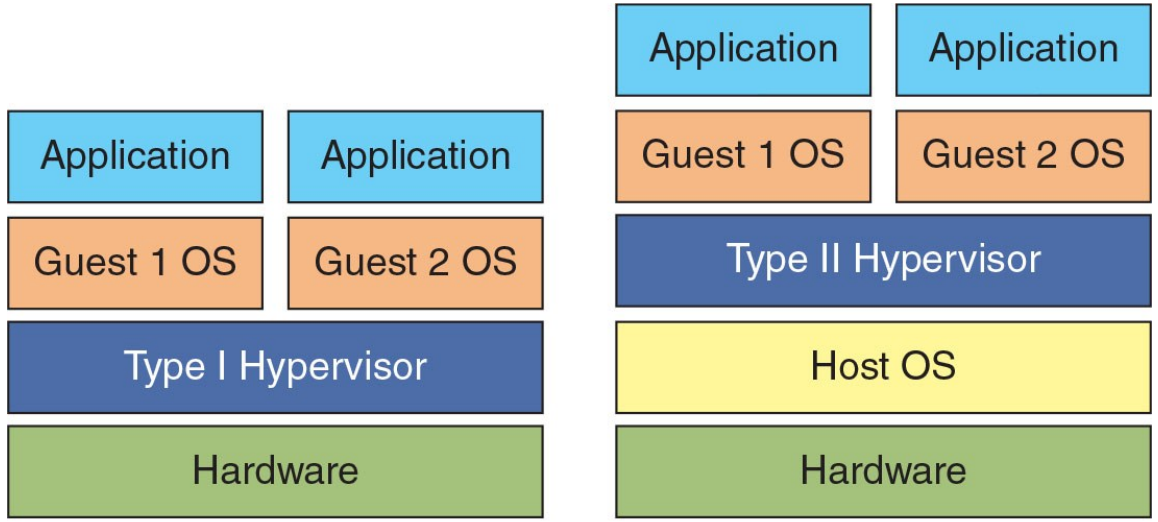


Figure 10-7 Type I and Type II hypervisors

Figure 10-7 Type I and Type II hypervisors

Defining Virtualization

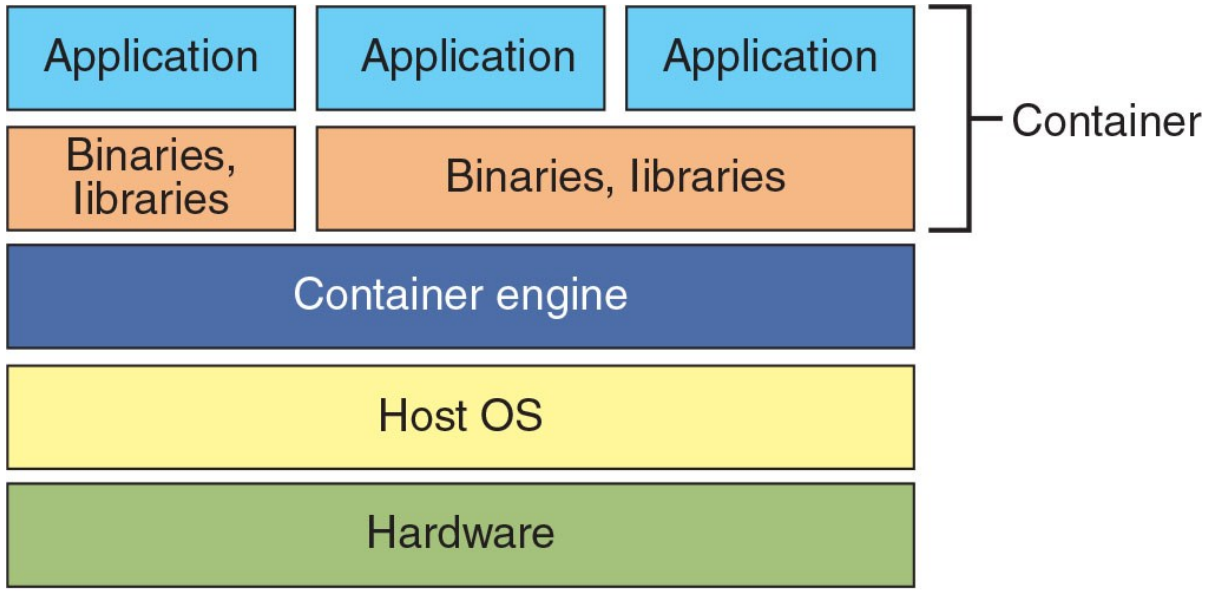


Figure 10-8 Container

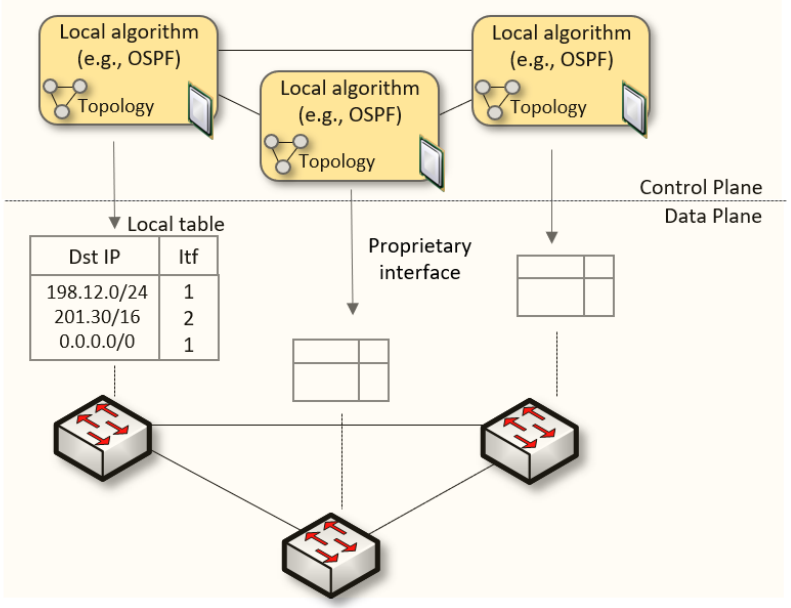
Figure 10-8 Container

Defining Virtualization

- Advantages of Virtualization
  - New virtual server machines can be made available (*host availability*) and resources can easily be expanded or contracted as needed (*host elasticity*)
  - Can reduce costs
    - Fewer physical computers must be purchased and maintained
  - Can provide uninterrupted server access to users
    - Supports *live migration* which allows a virtual machine to be moved to a different physical computer with no impact to users

Traditional (Legacy) Networking

- Since the explosive growth of the Internet in the 1990s, the networking industry has been dominated by closed and proprietary hardware and software
- The interface between control and data planes has been historically proprietary
  - Vendor dependence: slow product cycles of vendor equipment, no innovation from network owners
  - A router is a monolithic unit built and internally accessed by the manufacturer only

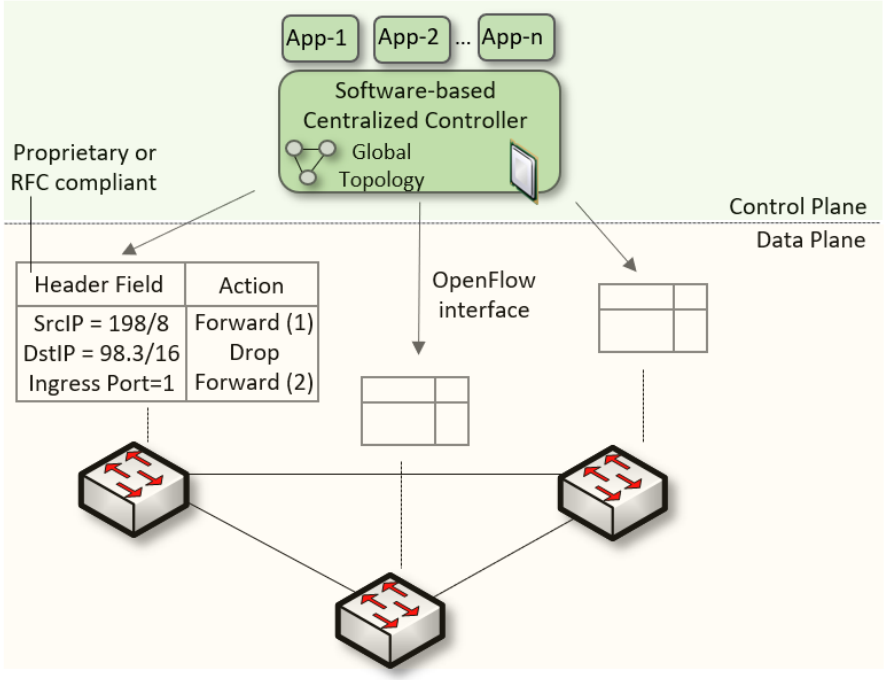


Infrastructure as Code

- **Software-defined network (SDN)**
  - An SDN virtualizes parts of the physical network so that it can be more quickly and easily reconfigured
  - This is accomplished by separating the *control plane* from the *data plane*
  - If traffic needs to flow through the network:
    - It receives permission from the SDN controller, which verifies the communication is permitted by the network policy of the enterprise
    - Once approved, the SDN controller computes a route for the flow to take and adds an entry for that flow in each of the switches along the path

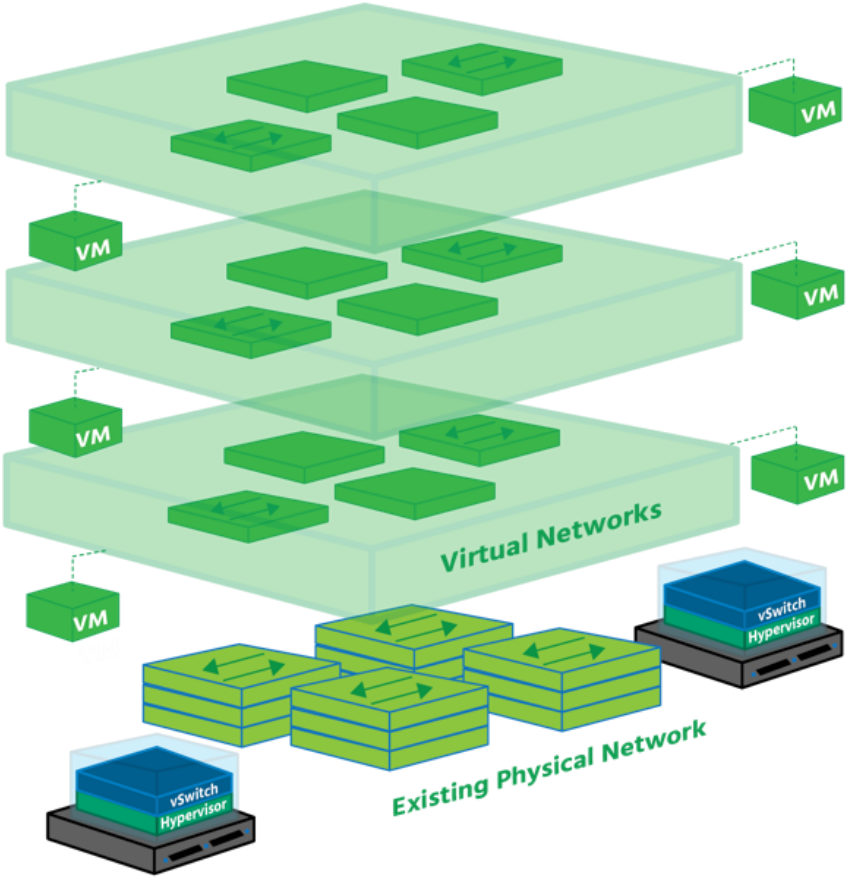
SDN

- Protocol ossification has been challenged first by SDN
- SDN (1) explicitly separates the control and data planes, and (2) enables the control plane intelligence to be implemented as a software outside the switches
- The function of populating the forwarding table is now performed by the controller



Virtual Networks in Physical Networks

- Virtual networks dramatically increase the scope of physical networks
- Virtual networks can run in isolation along side or on top of identical physical networks
- Each network is unaffected by the events on another network



Infrastructure as Code

- Software-Defined Visibility (SDV)
  - **Software-defined visibility (SDV)** is a framework that allows users to create programs in which critical security functions can be automated
  - SDV allows network administrators to automate multiple functions in a network infrastructure including:
    - Dynamic response to detected threat patterns
    - Adjustments to traffic mode configurations for in-line security tools
    - Additional IT operations-management functions and capabilities

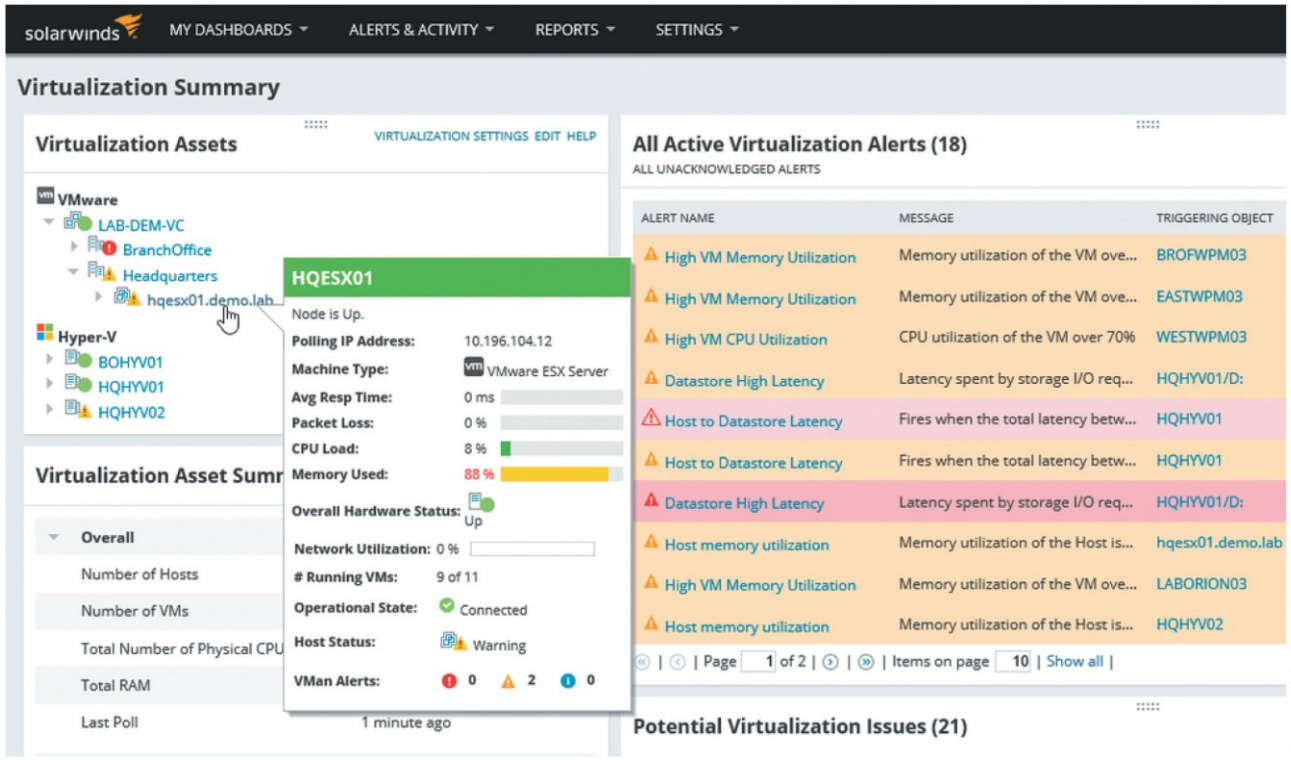
Security Concerns for Virtual Environments

- Security-related advantages of virtualization:
  - Test latest security updates by downloading on a virtual machine before installing on production computers
  - A *snapshot* of a particular state of a virtual machine can be saved for later use
  - Testing the existing security configuration *(security control testing)* can be performed using a simulated network environment
  - VMs can promote security segregation and isolation
  - A suspicious program can be loaded into an isolated virtual machine and executed *(sandboxing)*
    - If the program is malware, only the virtual machine will be impacted

Security Concerns for Virtual Environments

- Security concerns for virtualized environments:
  - Not all hypervisors have the necessary security controls to keep out attackers
  - Existing security tools were designed for single physical servers
  - VMs must be protected from both outside networks and other VMs on the same physical computer
  - VMs may be able to “escape” from the contained environment and directly interact with the host OS
    - Important to have **virtual machine escape protection**
- *Virtual machine sprawl* is the widespread proliferation of VMs without proper oversight or management
- Combating VM sprawl is called **virtual machine sprawl avoidance**
  - Installing a virtual machine manager can help

Security Concerns for Virtual Environments



Source: SolarWinds

Figure 10-10 Virtual machine manager

Figure 10-10 Virtual machine manager

Secure Network Protocols

- Common secure network protocols include:
  - Simple Network Management Protocol (SNMP)
  - Domain Name System (DNS) Security Extensions
  - File Transfer Protocol
  - Secure email protocols
  - Lightweight Directory Access Protocol (LDAP)
  - Internet Protocol version 6 (IPv6)

Simple Network Management Protocol (SNMP)

- SNMP is used to manage network equipment and is supported by most network equipment manufacturers
- It allows administrators to remotely monitor, manage, and configure network devices
- SNMP functions by exchanging management information between network devices
- Each SNMP-managed device has an agent or a service
  - Listens for and executes commands
- Agents are password protected
  - Password is known as a *community string*
- Security vulnerabilities were present in SMNP versions 1 and 2
  - Version 3 uses usernames and passwords along with encryption to address vulnerabilities

Domain Name System Security Extensions (DNSSEC)

- DNS is often the focus of attacks
  - DNS poisoning and DNS hijacking are examples
- These attacks can be thwarted by using **Domain Name System Security Extensions** (**DNSSEC**)
  - DNSSEC adds additional resource records and message header information which can be used to verify the requested data has not been altered in transmission
- Using asymmetric cryptography, a private key that is specific to a zone is used in encrypting a hash of a set of resource records
  - Which is then used to create the digital signature to be stored in the resource record

File Transfer Protocol

- File transfer protocol (FTP) is an unsecure protocol used to connect to an FTP server in order to transfer files
- Methods for using FTP on local host computer
  - *Using an FTP client*
  - *From a command prompt*
  - *Using a web browser*
- FTP vulnerabilities include:
  - FTP does not use encryption
  - Files transferred using FTP are vulnerable to man-in-the-middle attacks

File Transfer Protocol

- There are two options for secure transmissions over FTP
  - FTP Secure (FTPS) uses SSL or TLS to encrypt commands sent over the control port (port 21)
    - The data port (port 20) may not be encrypted
  - Secure FTP (SFTP)
    - Uses only a single TCP port instead of two ports
    - All data and commands are encrypted

Secure Email Protocols

- Earlier email systems use two TCP/IP protocols to send and receive messages:
  - Simple Mail Transfer Protocol (SMTP) and Post Office Protocol (POP)
- IMAP (Internet Mail Access Protocol) is a more recent and advanced email system
- As a means of security, a *mail gateway* monitors emails for unwanted content and prevents these messages from being delivered
- A mail gateway can automatically and transparently encrypt outbound email messages

Lightweight Directory Access Protocol (LDAP)

- A **directory service** is a database stored on the network that contains information about users and network devices
  - The directory service also keeps track of all the resources on the network and a user's privileges to those resources and grants or denies access based on the directory service information
- **Lightweight Directory Access Protocol** (**LDAP**) makes it possible for almost any application running on any computer platform to obtain directory information
- A weakness of LDAP is that it can be subject to **LDAP injection attacks**
  - This may allow an attacker to construct LDAP statements based on user input statements
- The defense against LDAP injection attacks is to examine all user input before processing

Internet Protocol Version 6 (IPv6)

- IPv6 addresses weaknesses of IPv4 and also provides other significant improvements
  - IPv6 increases the number of available addresses
- IPv6 has enhanced security features:
  - IPv6 can implement end-to-end encryption
    - This makes man-in-the-middle attacks significantly more difficult
  - IPv6 supports more secure name resolution
    - The Secure Neighbor Discovery (SEND) protocol can send cryptographic confirmation that an endpoint is who it claims to be
    - This makes ARP poisoning more difficult